



MLPC 2026

Data Exploration: Conclusion

Team Temporal Frameworks
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Biases - what needs to be considered?

- **DEVICE / HARDWARE bias:**
> 50 % iPhone recordings - might not generalise well on other microphones
- **DEMOGRAPHIC bias:**
exclusively collected by university students with similar acoustic environments
- **RECORDING CONDITION bias:**
staged, close-up recordings with unnaturally high SNR - will perform worse in real-life scenarios



Conclusion - is the dataset suitable for training?

- Dataset is **usable but challenging**
- High **feature correlation** → redundant information, harder model training
- Classes are **separable** in feature space, **but not cleanly**
- **Annotator disagreement** present, especially for ambiguous classes — likely to be mitigated by annotation review



Broader impact of publishing the dataset

POSITIVE:

- Enables **assistive tech** for **hearing-impaired** (smoke alarms, doorbells)
- Advances reproducible audio **AI research**

RISKS:

- Covert home surveillance
- Behavioral profiling / inferring **daily routines**

MITIGATION:

- Restrictive **license** and datasheet with **intended & prohibited use cases**
- Release **features only**, not raw audio